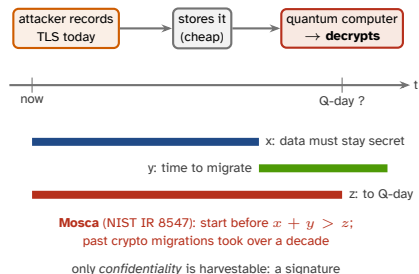
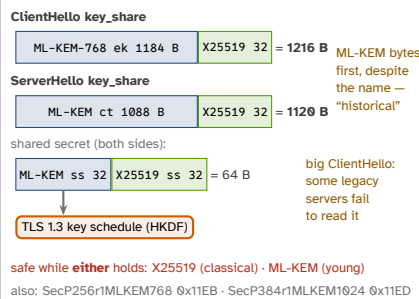


**In one line:** A large quantum computer running **Shor's algorithm** breaks every public-key primitive in use — RSA, (EC)DH, X25519, ECDSA, EdDSA — at any key size; symmetric ciphers and hashes stand. Traffic recorded *today* decrypts *then* (**harvest now, decrypt later**), so **key exchange** moved first: hybrid **X25519MLKEM768**, default in Chrome, Firefox, iOS 26, OpenSSL and OpenSSH. Since the 2026 estimates, **signatures** are urgent too: Google and Cloudflare now target 2029 for their whole migration.

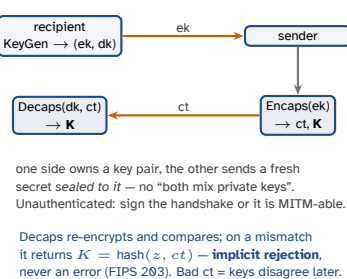
Harvest now, decrypt later



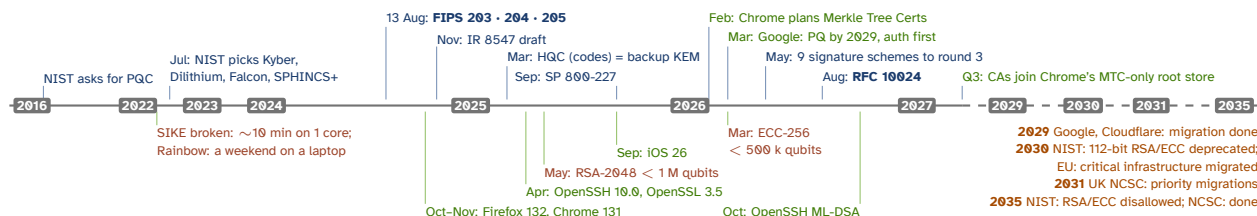
X25519MLKEM768 (RFC 10024, 0x11EC)



A KEM is not Diffie-Hellman



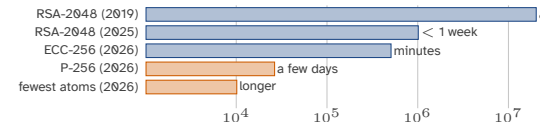
Timeline — standards and policy above, attacks and deployments below



What a quantum computer breaks

| primitive                             | attack                                             | consequence                                                             |
|---------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------|
| RSA, (EC)DH, X25519, ECDSA, EdDSA     | Shor (1994): factoring + discrete log in poly time | <b>broken at any key size</b> → replace with ML-KEM, ML-DSA, SLH-DSA    |
| AES-128                               | Grover (1996): $O(\sqrt{N})$ search                | NIST: still category 1, no transition planned; Apple: "move to AES-256" |
| AES-256, ChaCha20, SHA-2, SHA-3, HMAC | Grover                                             | stand; SHA-256 collision = category 2                                   |

Physical qubits to run Shor (log scale)



Blue: superconducting, 0.1% error (2019: 20 M); orange: neutral atoms. ECC-256: < 1200 logical qubits, minutes; RSA-2048: a week — **X25519 and P-256 fall first**. Google proved its circuit by zero-knowledge proof and kept it secret.

Why key exchange first — and why hybrid

- Harvested traffic is safe only if its key was agreed post-quantum *at the time*. A signature need only hold when checked — but forged firmware and CA signatures work for ever: after the 2026 estimates Google ranks authentication first (Android 17: ML-DSA).
- Web PKI: no post-quantum X.509 in Chrome's root store — **Merkle Tree Certificates**: a CA signs one tree head, a site sends a short inclusion proof.
- Hybrid**: secure if *either* part holds; new schemes do fall (SIKE, Rainbow). NIST: hybrid is temporary, a second move to PQ-only follows. CNSA 2.0 (NSA) already wants pure ML-KEM-1024 + ML-DSA-87, LMS/XMSS for firmware (*unverified*).

The NIST algorithms (sizes in bytes)

| standard                           | basis                         | sets → category · sizes                                                                                                                   |
|------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ML-KEM</b> FIPS 203 (Kyber)     | module lattice, $q = 3329$    | 512 / <b>768</b> / 1024 → 1 / 3 / 5; ek 800 / 1184 / 1568, ct 768 / 1088 / 1568, dk 1632 / 2400 / 3168, K 32. 768 = NIST default          |
| <b>ML-DSA</b> FIPS 204 (Dilithium) | module lattice, $q = 8380417$ | 44 / 65 / 87 → 2 / 3 / 5; pk 1312 / 1952 / 2592, sig 2420 / 3309 / 4627; sk 2560–4896 or a 32-B seed                                      |
| <b>SLH-DSA</b> FIPS 205 (SPHINCS+) | hashes only (SHA2 / SHAKE)    | 12 sets, 128 / 192 / 256 → 1 / 3 / 5; pk 32 / 48 / 64; sig s                                                                              |
| LMS, XMSS (SP 800-208)             | hashes, stateful              | 7856–29792, f (fast) 17088–49856; ≤ 2 <sup>64</sup> sigs per key reuse a one-time key = broken: state must never roll back; firmware only |
| FN-DSA · HQC                       | lattice · codes               | Falcon: draft FIPS 206 not public; HQC (backup KEM): draft due 2026, not out                                                              |

Categories: 1 = AES-128 key search · 2 = SHA-256 collision · 3 = AES-192 · 4 = SHA3-384 collision · 5 = AES-256. X25519 key 32, Ed25519 signature 64; ML-DSA-44's is 38 × that. Backup signatures, round 3: FAEST, HAWK, MAYO, MQOM, QR-UOV, SdItH, SNOVA, SQISign, UOV.

Who ships it — defaults and APIs

| where            | what                                                                                                                             |
|------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Chrome 131       | X25519MLKEM768; Kyber draft codepoint 0x6399 dropped                                                                             |
| Firefox 132      | mlkem768x25519 in TLS 1.3 (desktop)                                                                                              |
| Apple 26         | URLSession + Network.framework TLS; native VPN + IKEv2 API; SSH (macOS); iPhone ↔ Watch                                          |
| OpenSSL 3.5      | default keyshares X25519MLKEM768 + X25519; ML-KEM, ML-DSA, SLH-DSA                                                               |
| Go 1.24 · JDK 24 | crypto/tls default + crypto/mlkem · ML-KEM KEM API (JEP 496)                                                                     |
| OpenSSH          | 9.0: sntrup761x25519 · 10.0: mlkem768x25519-sha256 · 10.1: warns on non-PQ key agreement · 10.6: ssh-ml-dsa44-ed25519 signatures |
| Signal           | PQXDH (Sep 2023, Kyber in the handshake) → SPQR (Oct 2025) + Double Ratchet = "Triple Ratchet"                                   |
| iMessage PQ3     | IOS 17.4 (2024); Kyber-1024 + ECC; PQ rekey ~every 50 messages, ≤ 7 days                                                         |
| Cloudflare       | Sep 2026: ~70% of browser traffic PQ, only ~15% of origins                                                                       |

Apple platforms

- TLS**: the ClientHello offers X25519MLKEM768 + a key share; the *server* chooses, else classical — nothing to code, the server/CDN must support it. Check with `nscur l --tls-diagnostics <url>` (macOS 26). **Secure Transport** and own TLS stacks do not get it.
- CryptoKit** 26: MLKEM768, MLKEM1024, MLDSA65, MLDSA87, their SecureEnclave variants, HPKE XWingMLKEM768X25519\_SHA256\_AES\_GCM\_256; core formally verified against FIPS 203; same API server-side in Swift Crypto. 27: MLKEM768.OneTimePrivateKey — consumed by decapsulate, faster, not storable, not in the Secure Enclave.
- X-Wing** = ML-KEM-768 + X25519 as one KEM: pk 1216, ct 1120, private key a 32-B seed;  $K = \text{SHA3-256}(K_M \parallel K_X \parallel ct_X \parallel pk_X \parallel \text{label})$ .

Migration — crypto agility

- Inventory** every RSA/ECC use: TLS, VPN, SSH, custom protocols, wrapped keys, tokens, signatures, pinned keys, firmware roots (a crypto bill of materials).
- Rank by **secrecy lifetime** (Mosca's  $x$ ) and by **unpatchable verifiers**.
- Ride platform stacks that upgrade themselves (URLSession, OpenSSL, Go) — hand-rolled protocols move to HPKE with X-Wing.
- Make algorithms **negotiable**: IDs on the wire, no hard-coded key or signature lengths, rotation paths for keys and pins; budget kilobytes for keys, tokens, QR codes.

Traps

- ! "No quantum computer yet, so no rush" — today's traffic is the harvest;  $x + y > z$ .
- ! "Swap ECDH for ML-KEM" — deploy hybrid; and a bare KEM authenticates nobody.
- ! "Signatures can wait" — not for roots baked into hardware; estimates keep falling.
- ! Servers and middleboxes that cannot read a large ClientHello — test the fleet.
- ! A bad ciphertext raises no error — implicit rejection just yields a different key.

Remember

**Shor kills public key, Grover barely dents symmetric. Hybrid KEM now, PQ signatures next, deadlines 2029–2035 — and build for the next swap.**